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One-Step Multi-View Graph Clustering via Bottom-Up Structural Learning

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Abstract

In recent years, tensor-based methods have seen considerable success in multi-view clustering. However, the current approach has several limitations: 1) Insufficient exploration of underlying similarity information (i.e. latent representation); 2) Insufficient exploration of higher-order structure information of both inter-view and intra-view; 3) Treating clustering learning independently from tensor learning and the overall learning framework. To address these issues, we propose a unified framework called Bottom-up Structural Exploration for One-step Multi-view Graph Clustering (BSE_OMGC). Specifically, we first employ an anchor strategy to build similarity graphs, reducing the complexity of graph learning. To deeply represent the underlying similar information of the data and mitigate the influence of noise on similar structures in the original space, BSE_OMGC adaptively separates the noise matrix from the similarity graphs to learn high-quality enhanced graphs. Subsequently, from the bottom up, the enhanced graphs serve as the foundation for constructing high-order tensors. We rotate the constructed tensors and apply the t-TNN to preserve the low-rank properties and to better capture higher-order structure information of both inter-view and intra-view. Finally, we introduce a symmetric non-negative matrix factorization-based graph partitioning technique, which learns non-negative embeddings dur-

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ing dynamic optimization to reveal clustering results. This approach unifies clustering learning within the entire learning framework. Extensive experiments on multiple real-world multi-view datasets, along with comparisons to state-of-the-art methods, demonstrate the effectiveness and robustness of the proposed approach.

Keywords: Multi-view learning, Anchor graph, Low-rank tensor, One-step learning

1. Introduction

With the rapid emergence of large volumes of multi-view data [1], grouping unlabeled data into different clusters has become a highly researched topic among scholars. In practical scenarios, multi-view data come from various sources or distinct feature descriptions [2]. Such data naturally arise in a wide range of real-world applications, including multimedia analysis [3] (e.g., images described by visual, textual, and audio features), bioinformatics [4] (e.g., patient or gene clustering based on multi-omics data), social network analysis, and recommendation systems. In these applications, feature extraction techniques are commonly employed to obtain latent representations for each view [5], aiming to reduce noise and redundant information. However, relying solely on feature-level representations is often insufficient to fully characterize the intrinsic relationships among samples. As a result, it becomes necessary to further model and integrate the structural relationships between samples across different views.

Graph-based learning methods offer high flexibility and provide an effective means to explicitly model such inter-sample relationships by naturally representing the complex relationships and structural characteristics among data points [6]. In recent years, most graph-based multi-view clustering methods have achieved satisfactory results [7]. These methods typically involve two main steps: 1) Learning high-quality similarity graphs from multiple views to represent the data's underlying similarity information; 2) Applying graph partitioning techniques (such as K-means and rotational spectral clustering) on these similarity graphs to derive clustering results. The first step is crucial as the clustering results depend on the quality of the learned similarity graphs. Existing techniques for learning similarity graphs can be categorized into predefined similarity graph learning [8], subspace similarity graph learning [9], and adaptive similarity graph learning [10]. The subspace-

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based and adaptive-based methods offer greater flexibility compared to predefined similarity graph learning. The predefined-based methods statically learns a fixed graph that can be partitioned, whereas the latter two can dynamically update the graph during iterative optimization. Although these methods efficiently integrate multi-view data, uncover underlying structural information, and learn more refined graphs for clustering, they often overlook spatial structural information embedded within the graphs. To address this, researchers initially considered using graph regularization techniques to capture the global structural information of the data [11, 12], though this is limited to two-dimensional space. To explore higher-order relationships, tensor-based methods have been extensively employed in multi-view clustering [13, 14]. Researchers treat the similarity graph learned from each view as a slice of a tensor, constructing high-order tensors and imposing low-rank tensor constraints on them. For instance, Xia et al.[15] introduced the tensor based bipartite graph method, which leverages the spatial structure of the bipartite graph by minimizing the tensor Schatten-p norm. Similarly, Shu et al. [16] utilized the Schatten-p norm to explore the spatial structure and mutual information across views. Notably, tensor nuclear norm (t-TNN) based on singular value decomposition have been mathematically proven to achieve low-rank approximations, thereby capturing higher-order relationships between views more effectively [17, 18]. Consequently, numerous t-TNN-based multi-view graph learning methods have been proposed. For example, Wu et al.[19] proposed a unified graph and low-rank tensor learning method, which overlays the obtained adjacency matrices as third-order tensors through graph learning, and integrating the learning of adjacency matrices and low-rank tensor learning for joint optimization. Similarly, Chen et al. [20] stacked the tensor by adjacency matrices based on graph proximity learning. Moreover, Li et al. [21] introduced the Geman McClure loss function to enhance the learning of adjacency matrices for against noise and outliers.

Unfortunately, despite the promising results of the aforementioned methods, they still exhibit the following shortcomings: 1) In leveraging graphs to represent the underlying similarity information of data, these methods typically rely on either fixed graphs or dynamically learned graphs during optimization. Generally, a static, fixed graph can approximate the original data but is prone to being influenced by noise and outliers. Conversely, dynamically learned graphs can adjust during the learning process but are susceptible to the quality of the learned graph. Consequently, if the learned

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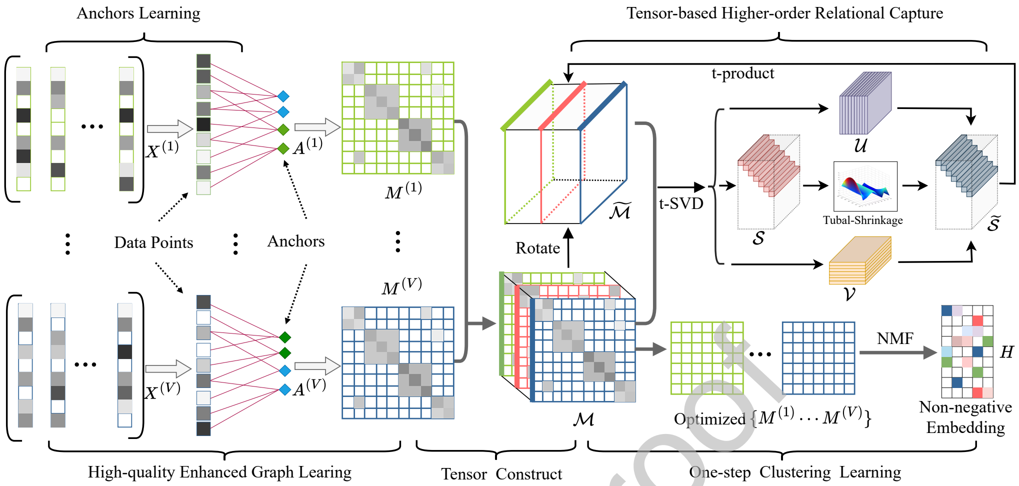

Figure 1: Overall framework of the proposed method.

graph is of low quality, it can adversely affect the subsequent tensor learning; 2) The graph structure learned from each view can be considered a slice of a tensor. When constructed along the view direction, this structure tends to capture higher-order relationships between views while overlooking those within a single view; 3) The learning of low-rank tensors and clustering indicator matrices is typically separated, ignoring their interrelationship and leading to suboptimal clustering results.

To address these issues, we propose a novel unified framework called Bottom-up Structure Exploration for One-step Multi-view Graph Clustering, as depicted in Figure 1. Specifically, for the "Bottom" layer structure exploration, we combine fixed and dynamic graphs. Firstly, we employ an anchor strategy to learn a fixed similarity graph. The anchor strategy not only reduces the computational complexity of graph learning but also preserves the overall data distribution. Subsequently, to mitigate the impact of noise in the original data and better represent the underlying structure, we dynamically separate a noise matrix from the similarity graph to learn an enhanced graph. For the "Up" layer higher-order structure exploration, we focus on both inter-view and intra-view relationships. For inter-view exploration, we stack the enhanced graphs along the view direction to form a tensor. For intra-view exploration, we rotate the original tensor along the sample direction, transforming the slice direction into inter-sample relationships. We then impose t-TNN constraints on both the original and rotated tensors to jointly capture higher-order structural information of both inter-view and intra-view. In our one-step multi-view graph clustering approach, we introduce symmetric non-negative matrix factorization to dynamically cut the enhanced graphs during the optimization process, obtaining non-negative embedding representations that directly reveal the clustering results without additional clustering operations. This process is unified within the entire learning framework, allowing for joint optimization with tensor learning and mutual reinforcement, thereby avoiding the suboptimal results that may arise from isolated clustering steps. Overall, the contributions of this study are as follows:

- This paper proposes a novel approach for constructing tensors through graph learning, combining both static and dynamic methods to use the learned enhanced graphs as the foundation for tensor construction.
- In this paper, we capture the common structure and differences of graph data by applying t-TNN constraints to both the original tensor and the rotation tensor. This idea takes into account inter-view and intra-view correlations.
- This paper introduces a one-step multi-view graph clustering method that utilizes symmetric non-negative matrix factorization to obtain clustering representations in a single step. This integration allows for joint optimization of clustering learning with tensor learning and the entire framework.

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2. Related Work

In this section, to facilitate the reader's understanding, we will introduce some related work to this paper, including spectral clustering in multi-view graph clustering, as well as the notation and definitions of tensors.

2.1. Spectral clustering

In spectral clustering, a similarity graph S is learned from the original data points $\{x_1, x_2, \cdots, x_n\}$. After obtaining the similar graph S, the relaxation objective function of spectral clustering using the graph normalization cutting method, which can be expressed as:

$$\min_{H^T DH} Tr(H^T LH) \tag{1}$$

where $L \in \mathbb{R}^{n\times n}$ is the Laplace matrix, expressed as $L = D - S$, the $D = diag(d_1, d_2, \cdots, d_n) \in \mathbb{R}^{n\times n}$ and the $I \in \mathbb{R}^{n\times n}$ represents the degree matrix and identity matrix, respectively. $H \in \mathbb{R}^{n\times k}$ represents the relaxed variable, k represents the top k smallest eigenvalues, which can be indicative of the number of clusters in spectral clustering. Then, let $F = D^{\frac{1}{2}}H$, the Eq.(1) can be restated as follow [22]:

$$\min_{F^TF=I} Tr(F^T D^{-\frac{1}{2}} L D^{-\frac{1}{2}} F) \tag{2}$$

Let's put $L = D - S$ into Eq.(2), and get the following equation:

$$\max_{F^TF} Tr(F^T D^{-\frac{1}{2}} S D^{-\frac{1}{2}} F) \tag{3}$$

The optimal solution of Eq.(3) comprises the k eigenvectors associated with the largest eigenvalues of the matrix $D^{-\frac{1}{2}} S D^{-\frac{1}{2}}$. Furthermore, matrix F denotes the spectral embedding of the original feature matrix, from which the final discrete clustering result can be derived through K-means clustering or spectral rotation.

2.2. Symmetric non-negative matrix factorization

In recent years, non-negative matrix factorization (NMF) method provides a low-rank approximation of a non-negative matrix [23]. The study of NMF computational methodologies are mentioned in [24] and many methods also has been successfully used as a clustering method [25]. However, it is still not a general clustering algorithm which can perform well in every circumstance. it assumes a linear combination of basis vectors to approximate the original data, which may not work well when the data has a non-linear structure or lies on a non-linear manifold [22].

However, symmetric non-negative matrix factorization is introduced as a more powerful alternative to NMF for clustering. Unlike NMF, which factorizes an arbitrary matrix, SymNMF works on a similarity matrix and allows for a more flexible approach to define the relationships between data points [26]. To be specific, mathematically, for a matrix $M \in \mathbb{R}^{n\times n}$, the goal of SymNMF can be represented as:

$$\min_{H} \|M - HH^T\|_F^2 \ \ s.t. H \geq 0 \tag{4}$$

where $H \in \mathbb{R}^{n\times K} \geq 0$ represents that all the elements are non-negative, and k is a pre-specified parameter. In addition, Eq.(4) can be used to deal with

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clustering problems via conducting an orthogonal constraint over H. It can be rewritten as the following formulation:

$$\min_{H} Tr(HH^T HH^T) - 2Tr(H^T MH) \quad (5)$$

Then, if we apply orthogonal constraints to H, denoted $HH^T = I$, and replace $D^{-\frac{1}{2}}SD^{-\frac{1}{2}}$ with M, Eq.(5) can be rewritten as:

$$\max_{H} Tr(F^T D^{-\frac{1}{2}} S D^{-\frac{1}{2}} F) \quad s.t \ HH^T = I \quad (6)$$

By revisiting the objective of relaxed Ncut, we observe that Eq.(6) aligns with the relaxed goal of Ncut as expressed in Eq.(3). This observation motivates us to integrate both methods into a unified framework in our study, which allows additional clustering operations to be integrated into tensor learning. We will explore this in more detail in Section 3.

2.3. Notations and Preliminaries

We use lower case letters (e.g. a) for vectors, upper case letters (e.g. A) for matrices, and calligraphy letters (e.g. $\mathcal{A}$) for tensors. For a matrix $A \in \mathbb{R}^{n_1 \times n_2}$, its Frobenius norm, nuclear norm, and $l_{2,1}$-norm are defined as $\|A\|_F = \sqrt{\sum_{ij} a_{ij}^2}$, $\|A\|_* = \sum_i \delta_i(A)$, $\|A\|_{2,1} = \sum_j \sqrt{\sum_i a_{ij}^2}$, respectively, where $\delta_i(A)$ is the i-th singular value of A. For a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, Frobenius norm, and infinity norm are defined as $\|A\|_1 = \sum_{ijk}$, $\|\mathcal{A}\|_F = \sqrt{\sum_{ijk} a_{ijk}^2}$, and $\|\mathcal{A}\|_\infty = \max_{ijk} |a_{ijk}|$, respectively. $\mathcal{A}(:,:,k)$, $\mathcal{A}(:,j,:)$, $\mathcal{A}(i,:,:)$ denote the i-th frontal, lateral, and horizantal slice, respectively. The transpose $\mathcal{A} \in \mathbb{R}^{n_2 \times n_1 \times n_3}$ is obtained by transposing each frontal slice and then reversing the order of transposed frontal slices through n_3. The fast Fourier transform (FFT) and corresponding inverse FFT operators of tensor $\mathcal{A}$ are $\mathcal{A}_f = fft((\mathcal{A}, [], 3))$ and $\mathcal{A} = ifft((\mathcal{A}_f, [], 3))$, respectively. The block vectorizing and corresponding inverse operation of $\mathcal{A}$ are $bvec(\mathcal{A}) = [A^{(1)}; A^{(2)}; \cdots; A(n_3)] \in \mathbb{R}^{n_1 n_3 \times n_2}$ and $fold(bvec(\mathcal{A})) = \mathcal{A}$. The block circulant matrix of tensor $\mathcal{A}$ is defind as $bcirc(\mathcal{A}) \in \mathbb{R}^{n_1 n_3 \times n_2 n_3}$ Frobenius norm, and infinity norm are defined as $\|A\|_1 = \sum_{ijk}$, $\|\mathcal{A}\|_F = \sqrt{\sum_{ijk} a_{ijk}^2}$, and $\|\mathcal{A}\|_\infty = \max_{ijk} |a_{ijk}|$, respectively. The transpose $\mathcal{A} \in \mathbb{R}^{n_2 \times n_1 \times n_3}$ is obtained by transposing each frontal slice and then reversing the order of transposed frontal slices through n_3.

Next, we will introduce several important definitions of tensors to help readers better understand.

Theorem 1. *(t-Product). The t-product of two tensors with matched dimension, $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$, is $\mathcal{A} * \mathcal{B} \in \mathbb{R}^{n_1 \times n_4 \times n_3}$.*

$$\mathcal{A} * \mathcal{B} = fold(bcirc(\mathcal{A})bvec(\mathcal{B})) \tag{7}$$

Theorem 2. *(Tensor Singular Value Decomposition (t-SVD) [27]). The t-SVD of $\mathcal{A}$ is defined as:*

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{8}$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$ and $\mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal, the tensor $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is f-diagonal.

Theorem 3. *(Tensor Nuclear Norm based on t-SVD (t-TNN) [28]). For the tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its tensor nuclear norm $\|\mathcal{A}\|_{\circledast}$ is defined as the sum of singular values with all the frontal slices of $\mathcal{A}_f$:*

$$\|\mathcal{A}\|_{\circledast} = \sum_{k=1}^{n_3} \|\mathcal{A}_f^{(k)}\|_{\circledast} = \sum_{i=1}^{\min(n_1,n_3)} \sum_{k=1}^{n_3} |S_f^{(k)}(i,i)| \tag{9}$$

where $S_f^{(k)(i,i)}$ is obtained by $M_f^{(k)} = U_f^{(k)} S_f^{(k)} (V_f^{(k)})^T$ in the Fourier domain.

3. The Proposed Method

In this section, we will introduce the proposed method of BSE_OMGC in detail. To enhance readability, we will primarily use symbols for clarity, as outlined in Table 1.

3.1. "Bottom" Structural Exploration

In the context of exploring underlying similarity information, as mentioned in the introduction, learning a high-quality similarity graph is crucial for multi-view learning. Therefore, we first adopt an anchor strategy to statically learn a fixed similarity graph A. Modeling sample similarities via anchors effectively constrains the similarity graph to a low-dimensional manifold spanned by a small number of representative anchors, where similarities are captured through shared anchor associations. This low-dimensional constraint suppresses spurious connections caused by noise and promotes smoothness and local consistency in graph structure. Specifically, we employ the K-means algorithm to obtain the anchors, as detailed below:

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Let $W = [w_1, w_2, \cdots, w_m]^T \in \mathbb{R}^{m\times d}$ represent the generated anchor matrix by K-means, which m is anchor number. And use $Z \in \mathbb{R}^{n\times m}$ measures the similarity between the original data point $X = [x_1, x_2, \cdots, x_n]^T \in \mathbb{R}^{n\times d}$ and generated anchors W. The (i, j)-th element of Z can be expressed as follows:

$$z_{ij} = \frac{K(x_i, w_j)}{\sum_{s\in\Phi_i} K(x_i, w_s)} \;\; \forall j \in \Phi_i \tag{10}$$

where $\Phi_i \in \{1, 2, \cdots, m\}$ represents the k-nearest neighbor index of X_i in W, $K(\cdot)$ is kernel function.

However, to avoid the extra parameters brought by the kernel. According to work [29], we adopt a parameter-free but effective neighbor assignment strategy, then the Eq.(10) can be rewritten as:

$$\min_{\mathbf{z}_i^T\mathbf{1}=1} \|\mathbf{x}_i - \mathbf{w}_j\|_2^2 + \gamma(z_{ij})^2 \tag{11}$$

where $\mathbf{z}_i^T$ is the i-th row of Z and the regularization parameter of γ can be set as $\gamma = \frac{k}{2}\|x_i - w_{k+1}\|_2^2 - \frac{1}{2}\sum_{h=1}^{k}\|x_i - w_h\|_2^2$, which k represents learning a sparse z_i with exactly k non-zero values. The Eq.(11) is solved by the following equation:

$$z_{ij} = \begin{cases} \frac{\|x_i - w_{k+1}\|_2^2 - \|x_i - w_j\|_2^2}{k\|x_i - w_{k+1}\|_2^2 - \sum_{h=1}^{k}\|x_i - w_h\|_2^2} & j \leq k \\ 0 & j \geq k \end{cases} \tag{12}$$

Then, based on the sparse metric matrix Z, similar graph $A \in \mathbb{R}^{n\times n}$ can be generated as following [30]:

$$A = Z\triangle^{-1}Z^T \tag{13}$$

where the diagonal matrix $\triangle \in \mathbb{R}^{n\times n}$ is defined as $\triangle_{ij} = \sum_{i=1}^{n} z_{ij}$. Based on the above explanations, it is evident that a notable reduction in time complexity can be achieved, as it involves the cost of $O(nm)$.

However, it is noteworthy that real-world data typically contain noise, which can cause the statically learned similarity graph A to be susceptible to outliers, resulting in an incomplete structure that fails to accurately represent the original data and capture the underlying structure. Therefore, to alleviate this problem, we employ adaptive graph learning to dynamically separate a set of sparse noise matrices $\{E^{(v)}\}_{v=1}^{V}$ from the similarity graph $A^{(v)}$ of each

Table 1: Notations and descriptions.

Notation	Description	Notation	Description
V	Total number of views	$X^{(v)}$	Original feature matrix
n	Total number of samples	$A^{(v)}$	Anchor graph
d	Feature dimension	$M^{(v)}$	Enhanced graph
v	v-th view	H	Non-negative embedding
k	Cluster number	$F^{(v)}$	Spectral embedding

view. This approach aims to obtain a more robust and high-quality enhanced graph $M^{(v)}$. The detailed process is as follows:

$$\begin{aligned} &\min_{E} \sum_{v=1}^{V} \|E^{(v)}\|_{2,1} \\ &s.t.\, A^{(v)} = M^{(v)} + E^{(v)}, v = 1, 2, \cdots, V \end{aligned} \tag{14}$$

where, $\|\cdot\|_{2,1}$ represents the $l_{2,1}$-norm, which effectively ensures the sparsity of the error matrix in its vectors.

To sum up, we have learned an enhanced graph $M^{(v)}$ to capture the underlying structure. However, most previous methods focus solely on the exploration of the underlying structure of graph representations in different views, neglecting the higher-order spatial information and structures of the graph. In the following sections, we will introduce methods to capture the hidden higher-order information.

3.2. "Up" Structural Exploration

For the exploration of higher-order structures, we expect to construct a third-order tensor from the enhanced graph and apply the t-TNN to capture the higher-order information hidden within the graph structures of each view. The detailed process is as follows:

$$\begin{aligned} &\min_{\mathcal{M},\mathcal{E}} \|\mathcal{M}\|_{\circledast} + \|\mathcal{E}\|_{2,1} \\ &s.t.\, A^{(v)} = M^{v)} + E^{(v)}, v = 1, 2, \cdots, V \\ &\mathcal{E} = [E^{(1)}; E^{(2)}; \cdots; E^{(V)}] \\ &\mathcal{M} = \Phi(M^{(1)}, M^{(2)}, \cdots M^{(V)}) \end{aligned} \tag{15}$$

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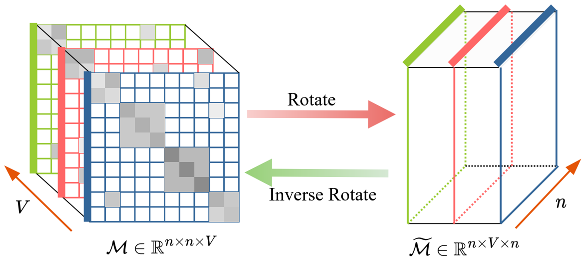

Figure 2: Illustration of tensor $\mathcal{M}$ and rotated tensor $\widetilde{\mathcal{M}}$. The integration of these two aspects facilitates a comprehensive examination of higher-order relationships both inter-view and intra-view.

where $\|\cdot\|_{\circledast}$ represents t-TNN, and the function of $\Phi(\cdot)$ represents the stacking of the enhanced graph $\{M^v\}_{v=1}^{V}$ into a third-order tensor $\mathcal{M}$.

However, for the problem Eq.(15), since the $\mathcal{M} \in \mathbb{R}^{n\times n\times V}$ stacking is performed along the view direction, the emphasis lies in capturing high-order relationships between views while inadvertently neglecting high-order relationships within views. To address this limitation, we introduce an additional rotation tensor denoted as $\widetilde{\mathcal{M}} \in \mathbb{R}^{n\times V\times n}$, achieved by rotating the tensor along the sample direction, as depicted in Figure 2. The t-TNN is defined as the sum of singular values from the frontal slice. Therefore, upon rotation, high-order relationships among samples (intra-view) can be explored more comprehensively. In summary, we restate the problem Eq.(15) by adding an additional rotation tensor $\widetilde{\mathcal{M}}$, as follows:

$$
\begin{aligned}
&\min_{\mathcal{M},\mathcal{E}} \|\mathcal{M}\|_{\circledast} + \alpha\|\widetilde{\mathcal{M}}\|_{\circledast} + \beta\|E\|_{2,1} \\
&s.t.\ A^{(v)} = M^{v)} + E^{(v)}, v = 1, 2, \cdots, V \\
&\mathcal{E} = [E^{(1)}; E^{(2)}; \cdots; E^{(V)}] \\
&\mathcal{M} = \Phi(M^{(1)}, M^{(2)}, \cdots M^{(V)}), \widetilde{\mathcal{M}} = \Psi(\mathcal{M})
\end{aligned}
\tag{16}
$$

where α and β are penalty factors, which α controls the weight of the high-order tensor structure modeling term within and across views, β regulates the strength of the tensor-based noise matrix constraint. Ψ representing the functions obtained $\widetilde{\mathcal{M}}$ by rotating $\mathcal{M}$ along the sample direction.

With model Eq.(16), we have addressed the shortcomings mentioned in the introduction. Bottom-up structural exploration enables the learned enhanced graph $M^{(v)}$ to not only retain the underlying structure of the data but

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also capture the spatial relationships hidden in the graph structure, specifically, the higher-order information of both inter-view and intra-view. Therefore, after obtaining a high-quality enhancement graph $M^{(v)}$, we then show how to use $M^{(v)}$ to obtain the final cluster representation.

3.3. One-step Multi-view Graph Clustering

After obtaining the optimal view-specific graph representations $M^{(v)}$ from Eq.(16), most existing methods typically adopt a two-step strategy: graph partitioning techniques are first applied to derive embedding representations, followed by an independent clustering procedure such as K-means or spectral rotation. In this paradigm, representation learning and clustering are decoupled, causing the clustering stage to be entirely dependent on the fixed embeddings. Such separation not only makes clustering sensitive to initialization and local optima, but also prevents clustering feedback from refining the learned tensor representations, potentially leading to suboptimal results.

To overcome this limitation, we introduce symmetric non-negative matrix factorization (SymNMF) into the tensor learning framework, enabling clustering learning to be jointly optimized with representation learning. As described in Section 2.2, the Eq.(6) aligns with the relaxed goal of Ncut as expressed in Eq.(3), indicating that both Ncut and SymNMF can be considered as relaxations to the model $\min_H \|M - HH^T\|_F^2$. To be specific, when the constraints are orthogonal, the model is equivalent to relaxed Ncut, and non-negativity is equivalent to SymNMF. Therefore, we integrate non-negative embedding and spectral embedding, extending it to multiple views, with the objective function expressed as follows:

$$\min_{H,F^v} \sum_{v=1}^{V} \|M^{(v)} - H(F^{(v)})^T\|_F^2 \quad s.t \; H \geq 0, F^{(v)}(F^{(v)})^T = I \tag{17}$$

where $H \in \mathbb{R}^{n\times k}$ and $F^{(v)} \in \mathbb{R}^{n\times k}$ denote non-negative embedding and spectral embedding, respectively. And since H is non-negative, the largest value in i-th row represents the clustering assignment of the i-th data point. Hence, the clustering representation can be directly derived in a single step from the non-negative embedding representation through the aforementioned objective function, eliminating the need for additional clustering operations in the context of spectral embedding. In summary, we embed Eq.(17) into tensor

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learning Eq.(16), obtaining the final objective function as follows:

$$\begin{aligned}
&\min_{\mathcal{M},\mathcal{E}} \|\mathcal{M}\|_{\circledast} + \alpha\|\widetilde{\mathcal{M}}\|_{\circledast} + \beta\|\mathcal{E}\|_{2,1} + \lambda\sum_{v=1}^{V}\|M^{(v)} - H\big(F^{(v)}\big)^T\|_F^2 \\
&s.t.\ A^{(v)} = M^{(v)} + E^{(v)}, H \geq 0, \\
&F^{(v)}\big(F^{(v)}\big)^T = I, \mathcal{E} = [E^{(1)}; E^{(2)}; \cdots; E^{(V)}] \\
&\mathcal{M} = \Phi(M^{(1)}, M^{(2)}, \cdots M^{(V)}), \widetilde{\mathcal{M}} = \Psi(\mathcal{M})
\end{aligned} \tag{18}$$

λ balances the clustering loss based on symmetric non-negative matrix factorization, which is a penalty factor.By embedding this clustering objective into the tensor learning model, clustering learning and tensor learning are jointly optimized within a unified objective function. The clustering loss explicitly regularizes the learned graph representations, while the progressively refined tensor representations provide more discriminative inputs for clustering. This tight coupling enables mutual reinforcement during optimization and effectively avoids the suboptimality induced by isolated clustering steps, ultimately leading to improved clustering performance.

3.4. Self-adaptive Hyperparameters

It is well known that the hyperparameters in a loss function play a crucial role in balancing the contributions of different loss components within a model. In Eq(18), three hyperparameters, α, β and λ are involved. In addition to relying on empirical tuning for their selection, we propose a novel paradigm for hyperparameter adjustment, termed a self-adaptive hyperparameter updating scheme based on loss gradient feedback. This strategy enables the automatic adjustment of key parameters according to the current model error, thereby enhancing the generalization capability and adaptability of the model. The resulting method is referred to as SA-BSE_OMGC (Self-Adaptive BSE_OMGC). To be specific, the update rule for each hyperparameter in the iteration of t is: is described as follows:

$$p^{(t+1)} = p^{(t)}(1 - \eta\frac{\partial L}{\partial p}) \tag{19}$$

where η is learning rate, we set it to 0.01, and p represents the hyperparameters α, β and λ. $\frac{\partial \mathcal{L}}{\partial p}$ is the gradient of the current loss to that parameter. when a hyperparameter exhibits a strong influence on the model loss (i.e., a

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large gradient), its value is appropriately decreased to reduce its dominance; conversely, when its influence is relatively weak (i.e., a small gradient), the parameter is maintained or adjusted slightly.

3.5. Optimization Process

Eq.(18) can be effectively solved by using the Alternating Direction Method of Multipliers (ADMM). Since the objective function involves non-smooth regularization terms, such as the tensor nuclear norm and the $l_{2,1}$ norm, directly solving the problem in a joint manner is challenging. ADMM alleviates this difficulty by splitting the variables and decomposing the original problem into a series of simpler subproblems, each involving a single variable update, which significantly reduces the optimization complexity.

Moreover, when the augmented Lagrangian function is convex with respect to each optimization variable, ADMM enjoys theoretical convergence guarantees, ensuring that the proposed optimization process converges to a stable and reliable solution. Therefore, we formulate the augmented Lagrangian function for Eq. (18) as follows:

$$\begin{aligned}
&\mathbf{L}(\{M^{(v)}\}_{v=1}^{V};\{E^{(v)}\}_{v=1}^{V};\{F^{(v)}\}_{v=1}^{V};\mathcal{P};\mathcal{Q};H)\\
&=\|\mathcal{P}\|_{\circledast}+\alpha\|\mathcal{Q}\|_{\circledast}+\lambda\sum_{v=1}^{V}\|M^{(v)}-H(F^{(v)})^{T}\|_F^2\\
&+\beta\|\mathcal{E}\|_{2,1}+\sum_{v=1}^{V}(\langle K^{(v)},A^{(v)}-M^{(v)}-E^{(v)}\rangle\\
&+\tfrac{\mu}{2}\|A^{(v)}-M^{(v)}-E^{(v)}\|_F^2)+\langle\mathcal{W}_1,\mathcal{M}-\mathcal{P}\rangle\\
&+\tfrac{\rho}{2}\|\mathcal{M}-\mathcal{P}\|_F^2+\langle\mathcal{W}_2,\mathcal{M}-\mathcal{Q}\rangle+\tfrac{\rho}{2}\|\widetilde{\mathcal{M}}-\mathcal{Q}\|_F^2
\end{aligned}\tag{20}$$

where $\mathcal{P}$ and $\mathcal{Q}$ is the adjunct tensor variable, the matrices $K^{(v)}$, the tensor W_1 and W_2 are Lagrange multipliers. μ and ρ are the penalty parameters, they can be adjusted by utilizing an adaptive updating strategy. Thus, we update various variables via the approaches presented below.

1) Fix $\mathcal{P}$, $\mathcal{Q}$, $E^{(v)}$, $F^{(v)}$, H to optimize $M^{(v)}$: By removing irrelevant items, the problem can be changed as below:

$$\begin{aligned}
&\min_{A^{(v)}}\lambda\|M^{(v)}-H(F^{(v)})^{T}\|_F^2+\langle K^{(v)},A^{(v)}-M^{(v)}\\
&-E^{(v)}\rangle+\tfrac{\mu}{2}\|A^{(v)}-M^{(v)}-E^{(v)}\|_F^2+\langle W_1^{(v)},\\
&A^{(v)}-P^{(v)}\rangle+\tfrac{\rho}{2}\|M^{(v)}-P^{(v)}\|_F^2+\langle W_2^{(v)},\\
&A^{(v)}-Q^{(v)}\rangle+\tfrac{\rho}{2}\|M^{(v)}-Q^{(v)}\|_F^2
\end{aligned}\tag{21}$$

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Setting the derivative of Eq.(21) with respect to $M^{(v)}$ to be zero, the solution of $M^{(v)}$can be obtained by

$$\begin{aligned} M^{(v)} = &\big((2\lambda + \mu + 2\rho)I\big)^{-1}\big(2\lambda H(F^{v})^{T} + \mu(A^{V} - E^{V}) \\ &+\rho(P^{(v)} + Q^{(v)}) + K^{(v)} - W_1^{(v)} - W_2^{(v)}\big) \end{aligned} \tag{22}$$

2) Fix $\mathcal{P}$, $\mathcal{Q}$, $M^{(v)}$, $F^{(v)}$, H to optimize $E^{(v)}$: the problem becomes:

$$\begin{aligned} &\min_{E^{(v)}} \beta\|E^{(v)}\|_{2,1} + \sum_{v=1}^{v}\big(\langle K^{(v)}, A^{(v)} - M^{(v)} \\ &-E^{(v)}\rangle + \tfrac{\mu}{2}\|A^{(v)} - M^{(v)} - E^{(v)}\|_F^2\big) \\ &= \min_{E^{(v)}} \beta\|E^{(v)}\|_{2,1} + \tfrac{\mu}{2}\|E^{(v)} - D\|_F^2 \end{aligned} \tag{23}$$

where $D = [D^{(1)}; D^{(2)}; \cdots D^{(V)}]$ is obtained by vertically concatenating $D^{(v)}$ along column and $D^{(v)} = A^{(v)} - M^{(v)} + \frac{1}{\mu}K^{(v)}$. Following the solution proposed in the literature [31], this minimization problem has the following solution:

$$E^{*}_{:,i} = \begin{cases} \frac{\|D_{:,i}\|_2 - \frac{\beta}{\mu}}{\|D_{:,i}\|_2} D_{:,i} & \frac{\beta}{\mu} \geq 0 \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

where $D_{:i}$ represents the i-th column in matrix D.

3) Fix $\mathcal{Q}$, $E^{(v)}$, $M^{(v)}$, $F^{(v)}$, H to optimize $\mathcal{P}$: the optimization of $\mathcal{P}$ is to address the following problem:

$$\begin{aligned} &\min_{\mathcal{P}} \|\mathcal{P}\|_{\circledast} + \langle \mathcal{W}_1, \mathcal{M} - \mathcal{P}\rangle + \tfrac{\rho}{2}\|\mathcal{M} - \mathcal{P}\|_F^2 \\ &= \min_{\mathcal{P}} \|\mathcal{P}\|_{\circledast} + \tfrac{\rho}{2}\|\mathcal{P} - (\mathcal{M} + \tfrac{1}{\rho}\mathcal{W}_1)\|_F^2 \end{aligned} \tag{25}$$

which is referred to as tensor multi-rank minimization and can be solved via the following process [32].

For a scalar τ and two tensor $\mathcal{T} \in \mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{O} \in \mathbb{R}^{n_1\times n_2\times n_3}$, there is a global optimal solution for the following problem:

$$\min_{\mathcal{T}} \tau\|\mathcal{T}\|_{\circledast} + \tfrac{1}{2}\|\mathcal{T} - \mathcal{O}\|_F^2 \tag{26}$$

which solution can be computed by tensor tubal-shrinkage operator:

$$\mathcal{T}^{*} = \mathcal{C}_{n_3\tau}(\mathcal{O}) = \mathcal{U} * \mathcal{C}_{n_3\tau}(\mathcal{S}) * \mathcal{V}^{T} \tag{27}$$

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where $\mathcal{Q} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ and $\mathcal{C}_{n_3\tau}(\mathcal{S}) = \mathcal{S} * \mathcal{J}$. $\mathcal{J}$ is an f-diagonal tensor whose diagonal element in the Fourier domain is $\mathcal{J}_f(i,i,j) = (1 - \frac{n_3\tau}{\mathcal{S}_f(i,i)^{(j)}})^+$.

4) Fix $\mathcal{P}$, $E^{(v)}$, $M^{(v)}$, $F^{(v)}$, H to optimize $\mathcal{Q}$: As with $\mathcal{P}$ above, and the optimization of $\mathcal{O}$ is to address the problem:

$$\begin{aligned} &\min_{\mathcal{Q}} \alpha \|\mathcal{Q}\|_{\circledast} + \langle \mathcal{W}_2, \mathcal{M} - \mathcal{Q} \rangle + \frac{\rho}{2} \|\mathcal{M} - \mathcal{Q}\|_F^2 \\ &= \min_{\mathcal{Q}} \alpha \|\mathcal{Q}\|_{\circledast} + \frac{\rho}{2} \|\mathcal{P} - (\mathcal{M} + \frac{1}{\rho}\mathcal{W}_2)\|_F^2 \end{aligned} \tag{28}$$

whose solution can be obtained to solve $\mathcal{P}$.

5) Fix $\mathcal{Q}$, $\mathcal{P}$, $E^{(v)}$, $M^{(v)}$, $F^{(v)}$ to optimize H: Keeping the other variables constant, and removing irrelevant terms, we update H by solving the following problem:

$$\min_{H} \sum_{v=1}^{V} \omega^{(v)} \|M^{(v)} - H(F^{(v)})^T\|_F^2 \tag{29}$$

where $\omega^{(v)} = \frac{1}{\|M^{(v)} - H(F^{(v)})^T\|_F^2}$. As $F^{(v)}\left(F^{(v)}\right)^T = I$. The Eq.(29) can reformulate the above equation as:

$$\min_{H} \|\sum_{v=1}^{V} \hat{\omega}^{(v)} H(F^{(v)}) - H\|_F^2 \tag{30}$$

where $\hat{\omega}^{(v)} = \frac{\hat{\omega}^{(v)}}{\sum_{v=1}^{V} \hat{\omega}^{(v)}}$. Let $B = \sum_{v=1}^{V} \hat{\omega}^{(v)} H(F^{(v)})$, the optimal solution of Eq.(30) becomes:

$$H = \max(0, B) \tag{31}$$

6) Fix $\mathcal{Q}$, $\mathcal{P}$, $E^{(v)}$, $M^{(v)}$, H to optimize $F^{(v)}$: To be specific, we optimize F^v by solving the following problem:

$$\min_{F^{(v)}} \sum_{v=1}^{V} \|M^{(v)} - H(F^{(v)})^T\|_F^2 \quad s.t. F^{(v)}\left(F^{(v)}\right)^T = I \tag{32}$$

We can find that the Eq.(32) is the famous orthogonal problem. So a close form solution for this problem is:

$$F^{(v)} = UV^T \tag{33}$$

where USV^T is the SVD of $\left(M^{(v)}\right)^T H$.

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7) Update the Lagrange multipliers $K^{(v)}$, $\mathcal{W}_1$ and $\mathcal{W}_2$, and non-negative parameters μ and ρ.

$$\begin{aligned}
K^{(v)} &= K^{(v)} + \mu(A^{(v)} - M^{(v)} - E^{(v)}) \\
\mathcal{W}_1 &= \mathcal{W}_1 + \rho(\mathcal{M} - \mathcal{P}) \\
\mathcal{W}_2 &= \mathcal{W}_2 + \rho(\widetilde{\mathcal{M}} - \mathcal{Q}) \\
\mu &= \min(\xi * \mu, \mu_{max}) \\
\rho &= \min(\xi * \rho, \rho_{max})
\end{aligned} \tag{34}$$

where ξ is a constant to adjust the convergence speed, μ_{max} and ρ_{max} represent the maximal values of μ and ρ.

Overall, two related methods are presented: BSE_OMGC and its enhanced variant SA-BSE_OMGC. BSE_OMGC employs three hyperparameters to balance different learning components; while proper tuning is crucial for performance, it may also introduce sensitivity across datasets. To alleviate this issue, SA-BSE_OMGC incorporates an adaptive hyperparameter adjustment mechanism based on gradient loss feedback, allowing automatic updates during optimization. As discussed in Section 4.2, the two methods exhibit complementary strengths on different datasets, while both consistently outperform all competing baselines overall. The main algorithmic procedure of SA-BSE_OMGC is summarized in Algorithm 1, whereas BSE_OMGC can be derived by removing the hyperparameter update step, i.e., deleting "Update α, β and γ by Eq. (19)" from Algorithm 1.

3.6. Complexity

Regarding the computational complexity of Algorithm 1, we make the following detailed analysis, which mainly includes the steps of tensor calculation and singular value decomposition. Firstly, the construction of anchor graphs A is $O(Vnm)$. Then, in the process of updating, it takes $O(Vn^2)$ to compute the $\mathcal{E}$. For the tensor $\mathcal{M}$, we need to compute the FFT and inverse FFT of the $n \times n \times V$ tensor, which takes $O(Vn^2 log(n))$, and $O(Vn^3)$ operations to compute SVD of $n \times n$ matrix in the Fourier domain. Thus, the complexity of updating $\mathcal{M}$ is $O(Vn^2 log(n) + Vn^3)$. Correspondingly, the complexity of the rotation tensor $\widetilde{\mathcal{M}}$ is $O(Vn^2 log(n) + V^2 n^2)$, which takes $O(Vn^2 log(n))$ operations to compute the FFT and inverse FFT of a tensor with $n \times V \times n$, and only takes $O(V^2 n^2)$ operations to calculate the SVD of $n \times V$ in the Fourier domain. Finally, the complexity of computing H

Algorithm 1 : The proposed method (SA-BSE_OMGC)

Input: Multi-view Data $\{X^{(v)}\}_{v=1}^{V}$, Similarity Graph $\{A^{(v)}\}_{v=1}^{V}$, and Penalty Factor α, β and λ.
Initialize $\mathcal{P} = \mathcal{Q} = \mathcal{W}_1 = \mathcal{W}_2 = 0$, $E^v = 0$, $K^v = 0$, $\xi = 2$, $\epsilon = 10^{-7}$, $\mu = \rho = 10^{-1}$, $\mu_{max} = \rho_{max} = 10^{10}$
repeat
  Update H by Eq.(31);
  for $v = 1$ **to** V **do**
    Update $F^{(v)}$ by Eq.(33);
    Update $M^{(v)}$ by Eq.(22);
  end for
  Update $\mathcal{P}$ by Eq.(25);
  Update $\mathcal{Q}$ by Eq.(28);
  for $v = 1$ **to** V **do**
    Update $K^{(v)}$ by Eq.(34);
  end for
  Update $\mathcal{W}_1$, $\mathcal{W}_2$, μ and ρ by Eq.(34);
  Update α, β and λ by Eq.(19);
until $\sum_{v=1}^{V} \|M^{(v)} - P^{(v)}\|_\infty \leq \epsilon$ or $\sum_{v=1}^{V} \|M^{(v)} - Q^{(v)}\|_\infty \leq \epsilon$;
Output: Consistent non-negative embedding H.

and $F^{(v)}$ is $O(Vn)$ and $O(Vnm^2)$, respectively. In addition, we have $V \ll n$, $m \ll n$ and $log(n) \ll n$. To this end, the complexity of Algorithm 1 is bounded by $O(Vn^3)$.

4. Experiment

In this section, we conduct experiments on eight widely-used multi-view datasets, comparing them with eight state-of-the-art MvC approaches in terms of six metrics. All experiments are implemented using the MATLAB R2018a environment, utilizing an Intel Core i7-11400 575CPU (2.90GHz) with 32GB RAM.

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Table 2: Summarization of datasets.

Dataset	Samples	View	Class	Dataset	Samples	View	Class
CESC	124	4	3	BBCSport	544	2	5
NGS	500	3	5	Prokaryotic	551	2	4
Scene	2688	4	8	ACM	3025	3	3
Cifar10	10000	4	10	ALOI-100	10800	4	100

4.1. Experimental Settings

We conduct experimental comparisons on several publicly available multi-view benchmark datasets, inculding ***CESC***[1], ***BBCSport***[2], ***NGs***[3], ***Prokaryotic***[33], ***Scene***[4], ***ACM***[34], ***Cifra10***[5], ***ALOI-100***[35]. The information of these datasets is summarized in Table 2. As observed, the number of samples, features, and categories in these datasets exhibits considerable variation, offering a robust platform for comparing the performance of different clustering algorithms. The clustering performance of all alogorithms is evaluated using six widely-used metrics: accuracy (ACC), normalized mutual information (NMI), adjusted rand index (ARI), F-score, precision and purity, where ACC and NMI are the most common metrics. In general, higher values of these metrics indicate better clustering performance. We compare the proposed framework with the following baselines:

- **Essential Tensor Learning for Multi-view Spectral Clustering (ETLMSC) [36]**. It utilizes Markov chain learning probability transition matrices to capture view information and construct tensors.
- **Low-rank Tensor Based Proximity Learning for Multi-view Clustering (LTBPL) [20]**. The algorithm stacks multiple affinity matrices into a low-rank constrained tensor to capture the higher-order correlations among views.
- **Unified Low-rank Tensor Learning and Spectral Embedding for Multi-view Subspace Clustering (ULTLSE) [37]**. ULTLSE

[1] `https://tcga-data.nci.nih.gov/tcga/`
[2] `http://mlg.ucd.ie/datasets/segment.html`
[3] `https://marineregions.org/`
[4] `https://datahub.io/machine-learning/scene`
[5] `http://www.cs.toronto.edu/kriz/cifar.html`

proposes a unified framework that utilizes spectral embedding to generate a label indicator matrix.

- **Unified One-step Multi-view Spectral Clustering (UOMvSC) [38]**. It combines the graphs and embedding matrices of different views to obtain a unified graph.
- **Efficient and Effective One-Step Multiview Clustering (EEOMVC) [39]**. It utilizes anchor graphs to generate latent partition representations, especially suitable for large-scale datasets.
- **Fast Multi-view Clustering via Ensembles: Towards scalability, superiority, and simplicity (FastMIC) [40]**. FastMICE introduces the concept of random view groups to capture diversified relationships among views, enabling efficient multi-stage fusion.
- **Inclusivity Induced Adaptive Graph Learning for Multi-view Clustering (IIAGL) [41]**. It leverages inclusivity-based graph learning to extract shared and complementary information.
- **Latent Information-Guided One-Step Multi-view Fuzzy Clustering Based on Cross-View Anchor Graph (OMVFC) [42]**. It introduces an effective cross-view anchor graph learning approach to construct the similarity matrix of multi-view data and extract latent information from the graph.

4.2. Experimental Results

We investigated the performance of our method compared to eight clustering algorithms, and the results are shown in Tables 3. The observations are as follows:

- On the eight datasets, the proposed BSE_OMGC demonstrates competitiveness across all six evaluation metrics. For example, on both the Scene and ACM datasets, RGTBL achieves the best clustering results. Compared to OMVFC, which also utilizes anchor graphs, BSE_OMGC exhibits superior clustering performance, underscoring its effectiveness in eliminating redundant information while preserving low-rank structures.

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Table 3: Clustering performance of different methods on eight datasets. The largest and second largest values are bolded and underlined respectively (%)

Datasets	ETL MSC	LT BPL	ULT LSE	UOM vSC	EEO MVC	Fast MICE	II AGL	OM VFC	BSE_ OMGC	SA-BSE _OMGC
					ACC					
CESC	40.56	60.48	58.06	60.48	66.13	71.11	78.22	53.23	83.06	**83.87**
BBCSport	**100.00**	**100.00**	99.82	53.31	60.66	44.08	78.38	72.43	**100.00**	**100.00**
NGs	67.30	98.90	97.60	29.80	34.80	46.12	97.60	45.20	99.00	**99.20**
Prokaryotic	66.80	68.60	60.07	57.53	60.25	57.45	51.72	40.47	**73.68**	67.33
Scene	62.30	62.35	82.22	70.76	61.24	70.52	63.16	56.85	**85.53**	68.45
ACM	69.57	65.22	93.24	62.64	57.19	68.27	74.94	64.83	95.74	**98.64**
Cifar10	47.52	45.00	38.29	55.62	47.86	56.11	49.41	48.42	83.28	**95.84**
ALOI-100	58.54	62.96	28.06	82.68	56.80	75.73	66.01	63.72	**85.84**	79.67
					NMI					
CESC	37.06	23.59	24.30	23.59	35.19	44.24	52.26	26.83	54.07	**56.18**
BBCSport	**100.00**	**100.00**	99.38	27.22	45.72	23.06	80.48	57.09	**100.00**	**100.00**
NGs	59.60	97.10	95.47	16.68	22.87	36.92	92.58	34.81	97.21	**97.62**
Prokaryotic	33.52	33.29	30.47	**37.70**	23.07	28.59	29.96	30.52	35.42	24.14
Scene	50.80	60.33	78.49	61.68	49.89	58.60	55.69	49.60	**79.10**	68.21
ACM	56.99	47.45	79.49	45.89	24.00	46.36	51.96	41.09	84.20	**93.86**
Cifar10	39.67	40.12	36.16	50.57	44.17	47.33	40.24	39.13	87.05	**93.22**
ALOI-100	75.59	65.43	65.40	87.52	73.32	84.35	79.02	77.43	**92.50**	88.66
					ARI					
CESC	20.31	21.33	22.85	21.33	29.66	38.80	48.55	14.22	55.42	**58.13**
BBCSport	**100.00**	**100.00**	99.77	17.24	26.40	12.88	78.81	49.74	**100.00**	**100.00**
NGs	52.20	**98.50**	96.52	12.39	6.79	21.98	94.08	16.74	97.50	98.00
Prokaryotic	39.23	32.15	24.22	30.40	22.17	21.22	23.06	35.69	**43.70**	37.80
Scene	43.25	50.44	71.03	49.61	35.51	49.90	46.88	38.55	**71.93**	53.55
ACM	49.20	49.30	83.15	37.22	26.13	41.94	48.82	34.27	87.80	**96.01**
Cifar10	28.99	28.76	26.86	30.93	23.41	34.31	29.08	39.91	76.76	**91.24**
ALOI-100	47.14	57.93	23.31	41.95	14.56	66.42	52.22	51.05	**79.42**	73.02
					F-score					
CESC	37.04	58.86	54.68	58.86	53.90	71.40	73.22	46.39	71.71	73.61
BBCSport	**100.00**	**100.00**	99.83	43.97	50.73	37.77	83.88	62.65	**100.00**	**100.00**
NGs	62.13	97.60	97.21	28.18	35.27	41.73	95.26	40.76	98.02	**98.40**
Prokaryotic	58.27	64.60	57.46	56.29	54.93	52.24	48.06	35.32	66.02	**66.90**
Scene	50.48	58.59	75.06	56.48	45.42	56.44	54.28	46.55	**75.62**	60.11
ACM	68.57	70.64	88.49	63.16	55.54	61.92	66.73	56.47	91.88	**97.34**
Cifar10	36.12	38.30	22.00	39.26	33.16	41.29	36.31	36.87	79.31	**92.12**
ALOI-100	47.71	59.62	24.55	42.77	15.51	66.78	52.76	51.60	**79.63**	73.30
					Precision					
CESC	37.12	44.09	47.35	44.09	55.89	60.83	68.00	44.20	70.10	**71.14**
BBCSport	**100.00**	**100.00**	99.83	32.47	36.60	48.30	83.58	58.19	**100.00**	**100.00**
NGs	65.18	97.60	97.17	21.16	22.92	58.26	95.23	28.03	98.00	**98.37**
Prokaryotic	61.36	53.25	51.65	40.41	44.48	63.12	47.06	44.32	**65.65**	56.37
Scene	50.33	45.25	68.39	52.63	37.58	58.04	49.44	45.41	**73.13**	54.03
ACM	59.98	55.99	88.69	57.00	46.54	63.91	63.63	55.78	91.67	**97.39**
Cifar10	35.91	28.58	23.52	32.52	25.82	44.11	35.61	34.27	72.07	**91.95**
ALOI-100	43.96	45.19	15.10	30.22	13.81	72.16	48.66	45.38	**76.74**	69.64
					Purity					
CESC	49.35	61.29	58.87	61.29	67.74	60.58	67.16	60.48	83.06	**83.87**
BBCSport	**100.00**	**100.00**	99.82	53.31	64.71	59.00	85.47	77.02	**100.00**	**100.00**
NGs	69.68	98.80	97.16	31.00	35.20	70.16	97.60	46.00	99.00	**99.20**
Prokaryotic	71.41	72.41	61.34	59.35	67.70	76.55	68.78	56.81	**77.50**	67.51
Scene	63.85	62.43	82.25	70.76	61.24	72.55	65.88	58.56	**85.53**	68.45
ACM	69.57	65.26	94.24	62.64	57.19	73.71	74.94	64.83	95.74	**98.64**
Cifar10	48.79	45.24	38.33	57.78	49.17	59.82	50.71	29.27	83.28	**95.84**
ALOI-100	60.96	64.31	29.04	83.37	58.22	83.71	69.60	65.00	**86.38**	80.97

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- For the BBCSport dataset, our BSE_OMGC, ETLMSC, and LTBPL all achieved a clustering performance of 1, validating the superiority of tensor low-rank. However, on the Prokaryotic dataset, BSE_OMGC outperformed these two methods in terms of ACC by 6.88% and 5.08%, respectively. These results demonstrate the effectiveness of BSE_OMGC in capturing high-order relationships between views through dual-tensor constraints.

- From Table 3, it is evident that BSE_OMGC outperforms other multi-view clustering algorithms. This is because ETLMSC and LTBPL, two tensor-based methods, struggle with large-scale datasets, ULTLSE performs poorly due to the label indicator matrix generated using spectral embedding, which may not adequately capture the underlying structure of the data. In contrast, our BSE_OMGC integrates symmetric non-negative matrix factorization, enabling direct extraction of clustering results from learned non-negative embeddings. This unique approach contributes to the demonstrated superiority of BSE_OMGC.

- In addition, we compared the clustering performance of the original method BSE_OMGC with the improved variant SA-BSE_OMGC, which incorporates an adaptive hyperparameter adjustment mechanism. Experimental results demonstrate that SA-BSE_OMGC outperforms the original approach on five datasets (CESC, BBCSport, NGs, ACM, and Cifar100) with particularly notable improvements on the ACM and Cifar100 datasets. However, a slight performance degradation is observed on Prokaryotic, Scene, and ALOI-100.Specifically, SA-BSE_OMGC introduces a self-adaptive hyperparameter tuning strategy based on loss gradient feedback, which dynamically adjusts key parameters according to the current model error. This enables more balanced weighting of loss components. The mechanism proves especially effective on datasets with complex structures or high feature heterogeneity (e.g., ACM and Cifar100), where it facilitates more accurate modeling of the intrinsic data structure. In contrast, for datasets characterized by unstable feature representations or significant inter-view discrepancies (e.g., Prokaryotic, Scene, and ALOI-100), fluctuations in gradient signals may lead to suboptimal parameter updates, thereby impairing clustering performance.In summary, SA-BSE_OMGC demonstrates superior adaptability and generalization across most datasets,

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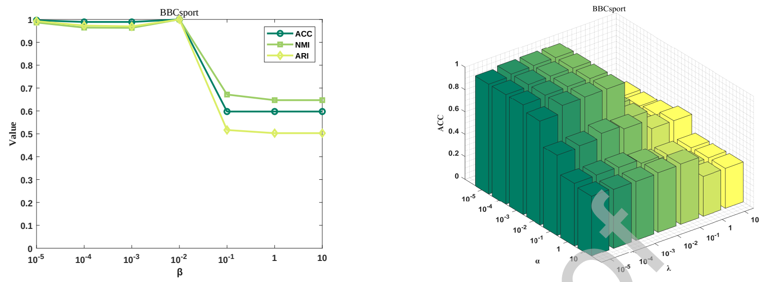
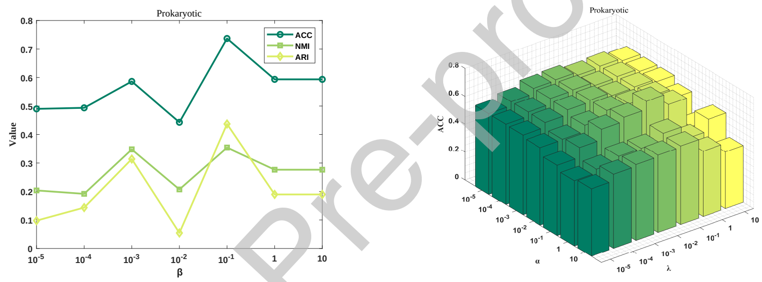
validating the effectiveness of incorporating adaptive hyperparameter optimization.



(a) BBCSport



(b) Prokaryotic

Figure 3: The parameter analysis of BBCSport and Prokaryotic for α, λ, and β.

4.3. Parameter Analysis

To further illustrate the effectiveness and robustness of our model, we conducted a comprehensive analysis of the parameters in the objective function. As shown in Figure 3 which include α, λ and β. The distinct values of these hyperparameters can have different effects on the algorithm's performance. In the experiment, we investigated the influence of β on clustering performance metrics ACC, NMI, and ARI while keeping other parameters fixed. We varied the parameter β within the range $[10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10]$ and plotted a line graph. Additionally, to further investigate the impact of α and λ on BSE_OMGC, the histogram in Figure 3 depicts the clustering performance metric ACC under different parameter settings of α and λ. Taking the BBCSport dataset as an example, the optimal clustering performance is achieved when $\alpha = 10^{-3}$, $\beta = 10^{-3}$ and $\lambda = 10^{-2}$, demonstrating

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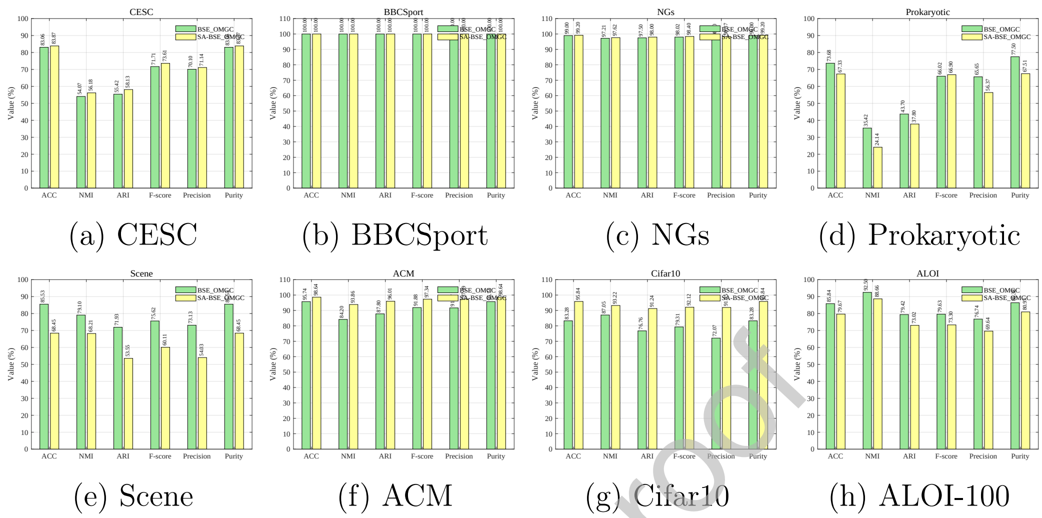

(a) CESC (b) BBCSport (c) NGs (d) Prokaryotic

(e) Scene (f) ACM (g) Cifar10 (h) ALOI-100

Figure 4: Comparison of empirically adjusted parameters (BSE_OMGC) and self-adaptive parameters (SA-BSE_OMGC) on eight datasets.

the effectiveness of this parameter configuration. Our method's clustering performance is less affected by parameter perturbations on the Prokaryotic datasets, but slightly more sensitive to the combination of these two parameters on the BBCSport datasets.

In order to alleviate the dependence of the model on hyperparameter sensitivity, we introduce an adaptive hyperparameter adjustment mechanism based on loss gradient feedback as described in Section 3.4. The mechanism automatically updates dynamically according to the influence of each hyperparameter on the total loss in the training process, so as to avoid the complexity caused by experience parameter adjustment to a certain extent. In order to verify the effectiveness of this mechanism, we drew bar charts of empirically adjusted parameters (BSE_OMGC) and self-adaptive parameters (SA-BSE_OMGC) for comparison, as shown in Figure 4, where the ordinate represents the accuracy rate ACC and the abscissa represents different data sets. As shown in Figure 4, the adaptive mechanism shows good adaptability and robustness on most data sets, especially on data sets with complex features or strong structural diversity.

4.4. Ablation

To verify the validity of rotation tensor and one-step multi-view graph clustering, we conducted cross-validation by systematically retaining one

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Table 4: Ablation experiments on eight datasets, the largest values are bolded.

Dataset	Method	ACC	NMI	ARI	F-score	Precision	Purity
CESC	BSE_OMGC	**83.06**	**54.07**	**55.42**	**71.71**	**70.10**	**83.06**
	BSE_OMGC-I	80.65	50.84	50.48	68.29	67.83	80.65
	BSE_OMGC-II	79.03	50.54	48.72	68.83	67.62	79.03
BBCSport	BSE_OMGC	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
	BSE_OMGC-I	88.60	80.95	81.83	86.21	85.42	88.60
	BSE_OMGC-II	97.06	91.02	91.99	93.88	94.98	97.06
NGs	BSE_OMGC	**99.00**	**97.21**	**97.50**	**98.02**	**98.00**	**99.00**
	BSE_OMGC-I	95.00	88.28	88.34	90.66	90.35	95.00
	BSE_OMGC-II	95.60	92.18	94.07	95.25	95.24	95.60
Prokaryotic	BSE_OMGC	**73.68**	**35.42**	**43.70**	**66.02**	**65.65**	**77.50**
	BSE_OMGC-I	66.43	32.08	34.20	56.38	65.94	72.23
	BSE_OMGC-II	60.07	20.51	10.12	53.63	43.67	66.24
Scene	BSE_OMGC	**85.53**	**79.10**	**71.93**	**75.62**	**73.13**	**85.53**
	BSE_OMGC-I	74.55	76.15	62.12	67.35	62.00	74.55
	BSE_OMGC-II	65.40	58.52	49.28	56.50	50.39	67.41
ACM	BSE_OMGC	**95.74**	**84.20**	**87.80**	**91.88**	**91.67**	**95.74**
	BSE_OMGC-I	87.50	66.26	67.60	78.44	78.20	87.50
	BSE_OMGC-II	89.69	66.12	71.86	81.28	81.03	89.69
Cifar100	BSE_OMGC	**83.28**	**87.05**	**76.76**	**79.31**	**72.07**	**83.28**
	BSE_OMGC-I	75.33	72.17	62.93	66.95	61.59	75.48
	BSE_OMGC-II	66.69	76.37	51.50	57.79	44.50	66.69
ALOI-100	BSE_OMGC	**85.84**	**92.50**	**79.42**	**79.63**	**76.74**	**86.38**
	BSE_OMGC-I	62.47	79.18	51.89	52.40	48.92	66.32
	BSE_OMGC-II	61.28	78.41	50.29	50.82	47.26	65.01

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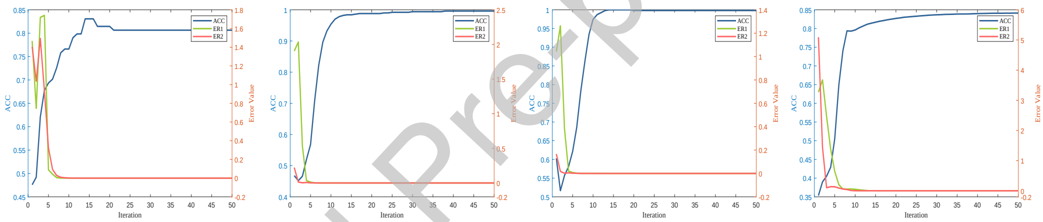
module while removing another. Firstly, we removed the one-step multi-view graph clustering module by eliminating matrix factorization term $\|M^{(v)} - H(F^{(v)})^T\|_F^2$, referred to as BSE_OMGC-I. Next, we removed the rotation tensor by excluding term $\widetilde{\mathcal{M}}$ from the objective function, referred to as BSE_OMGC-II. We conducted experiments on eight datasets, with results shown in Table 4. The results indicate that the model's performance significantly decreases when either of these modules is removed. This suggests that introducing the rotation tensor along the sample direction effectively captures higher-order intra-view relationships, thereby improving model efficiency. Furthermore, the one-step multi-view graph clustering achieved through matrix decomposition eliminates uncertainties in subsequent operations, enhancing clustering performance. Therefore, both the rotation tensor and the one-step multi-view graph clustering module have a positive impact on the model.



(a) CESC (c) NGs (b) BBCSport (d) Cifar

Figure 5: The convergence analysis of CESC, NGs, BBCSport and Cifar

4.5. Convergence

We plotted the error values of variables and clustering results with respect to the number of iterations. The errors of variables are defined as ER1=$\sum_{v=1}^{V} |M^{(v)} - P^{(v)}|\infty \leq \epsilon$ and ER2=$\sum_{v=1}^{V} |M^{(v)} - Q^{(v)}|_\infty \leq \epsilon$. From Figure 5, it can be observed that the errors monotonically decrease, and the algorithm typically converges within a few iterations. Additionally, to demonstrate the clustering performance of BSE_OMGC with iterations, we report the results of ACC at each iteration in Algorithm 1. It can be observed that the clustering performance of BSE_OMGC initially improves with the number of iterations and then stabilizes, effectively showcasing the effectiveness of our algorithm. These results clearly demonstrate the effectiveness and necessity of the learning process.

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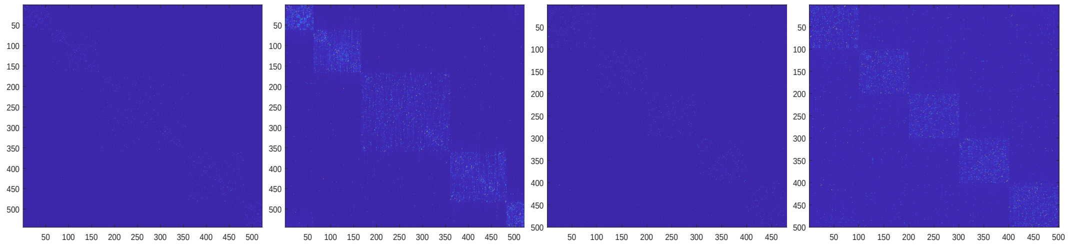
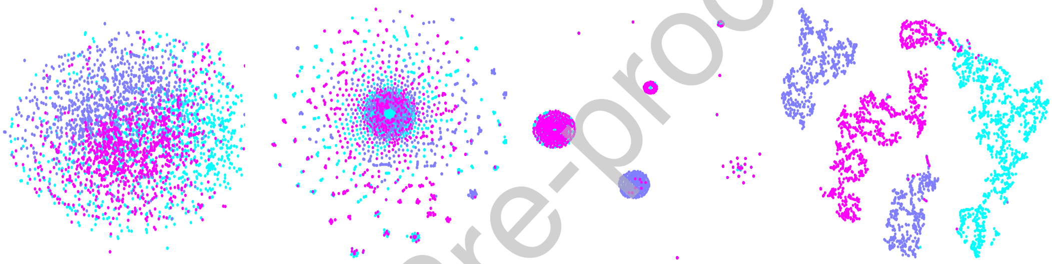
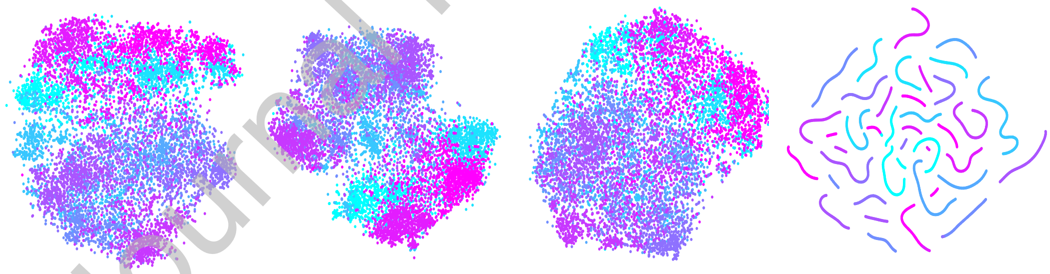


(a) BBCSport $A^{(1)}$ (b) BBCSport $M^{(1)}$ (c) NGs $A^{(1)}$ (d) NGs $M^{(1)}$

Figure 6: Visualization of the enhanced graph through Algorithm 1 on the BBCSport and NGs



(a) ACM $X^{(1)}$ (b) ACM $X^{(2)}$ (c) ACM $X^{(3)}$ (d) ACM H



(e) Cifar $X^{(1)}$ (f) Cifar $X^{(2)}$ (g) Cifar $X^{(3)}$ (h) Cifar H

Figure 7: t-SNE visualization of raw views $X^{(v)}$ and unified non-negative embedding representation H on ACM and Cifar datasets

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4.6. Visualization

In this section, we validate the model's ability to learn high-quality enhanced graphs that capture underlying structures. We also demonstrate its capability to reveal clustering results through non-negative embedding representations obtained via matrix decomposition techniques, utilizing various visualization methods.

Firstly, to assess the ability of high-quality enhanced graphs to capture underlying structures, we visualized the similarity graph $A^{(v)}$ and the enhanced graph $M^{(v)}$ for the BBCSport and NGs datasets. The results in Figure 6 demonstrate that, after appropriate iterations, our method effectively learns the similar structures of views while suppressing noise. Secondly, to evaluate the capability of non-negative embedding representations in revealing clustering results, we visualized the BBCSport and CIFAR datasets using t-SNE to display the unified non-negative embedding representations learned by our method. As shown in Figure 7, the unified non-negative embedding (Figure 7 d and 7 h) obtained by our method exhibit clearer and more compact cluster structures compared to the initial each view (Figure 7 a-c and 7 e-g), thereby underscoring the effectiveness of our approach in revealing clustering results.

5. Discussion

Several promising directions remain for future investigation based on the proposed model. First, at the architectural level, deep learning techniques—particularly attention-based encoder–decoder structures can be incorporated to further enhance multi-view graph representation learning, such as Transformers[43]. Owing to their strong capability in modeling complex features and cross-modal semantic relationships, Transformers have shown effectiveness in capturing high-order nonlinear dependencies. Future work may therefore explore integrating deep representation learning modules with the existing graph learning framework to improve its ability to model high-level semantic information and generalize to complex multi-view data.

Second, although multi-view clustering has achieved substantial performance improvements in recent years, most existing graph learning and clustering models still suffer from limited interpretability, rendering their internal decision-making processes difficult to understand. In practical applications, explaining why samples are assigned to specific clusters is therefore of critical importance. From this perspective, incorporating explainable artificial intelligence (XAI) mechanisms [44, 45] provides a promising pathway to enhance model transparency, for example through local explanation methods such as LIME [46], which characterize the relationships between model outputs and input features. Beyond general-purpose XAI techniques, recent studies have also explored interpretability directly within the multi-view clustering

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paradigm. For instance, Jiang et al. [47] proposed an interpretable multi-view clustering framework that leverages pseudo-labels to guide the construction and iterative refinement of explainable decision trees, thereby improving clustering transparency while maintaining competitive performance. Overall, systematically integrating interpretability mechanisms into multi-view clustering architectures remains a meaningful direction for future research.

Finally, from an application and deployment perspective, developing user-oriented interactive web applications or prototype systems is of practical significance [48]. By encapsulating complex multi-view clustering models into visual and interactive web-based tools, the usability and accessibility of the proposed method can be substantially improved [49]. Future work may therefore focus on integrating the proposed approach into lightweight web platforms that support clustering visualization and explanation analysis, facilitating its adoption in real-world multi-view data analysis and pattern recognition tasks.

6. Conclusion

This paper proposes a unified static–dynamic collaborative graph learning framework for multi-view clustering by integrating anchor-based static similarity graphs with dynamically learned graph structures through adaptive noise separation. By preserving global structural stability while flexibly capturing local structural variations, the proposed framework enables a fine-grained modeling of latent similarity relationships across multiple views. In addition, it effectively suppresses noise interference, leading to significantly improved discriminability and robustness of graph representations in scenarios characterized by noisy observations and highly heterogeneous feature distributions.

From a methodological perspective, the main contribution of this work lies not only in performance gains but also in the introduction of a general paradigm that jointly optimizes static–dynamic collaborative modeling and clustering. By tightly coupling graph representation learning and clustering learning within a unified optimization framework, the proposed approach avoids the error accumulation and suboptimal solutions commonly induced by conventional two-stage strategies that decouple graph construction from clustering. This representation–decision integrated modeling principle provides a reusable design guideline for multi-view clustering, graph structure learning, and related pattern recognition tasks, enabling researchers to more

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effectively coordinate structural modeling and downstream decision-making in complex structured data analysis.

Despite its effectiveness across a variety of multi-view clustering scenarios, the proposed method still exhibits certain limitations. In order to capture high-order correlations among multiple views, tensor-based modeling is employed, which enhances representational capacity at the cost of increased computational and storage complexity, particularly in large-scale or high-dimensional multi-view settings. Although an anchor-based graph construction strategy is adopted to alleviate this issue, achieving a more favorable balance between expressive power and computational efficiency under extreme-scale conditions remains an open problem. Future work may explore more efficient tensor decomposition or approximation techniques and, as discussed in Section 5, incorporate deep representation learning mechanisms to further enhance the modeling of complex nonlinear structures and improve scalability, thereby facilitating the application of the proposed framework to complex real-world multi-view problems, such as cancer subtype identification based on heterogeneous multi-omics data.

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CRediT authorship contribution statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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